

Sai Teja Reddy Jillella

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Professional Summary

Results-driven Full Stack Software Developer with 3+ years of experience architecting engaging front-end interfaces and scalable back-end systems. Skilled in React, Node.js, Python, Flask, and REST API design, with proven success in deploying secure applications on AWS and GCP. Demonstrated ability to streamline development through Docker, Kubernetes, and CI/CD pipelines while ensuring high performance and reliability.

Education

University of Cincinnati – Cincinnati, OH

May 2025

Master of Engineering, Computer Science

• **GPA:** 3.87/4.0

Professional Experience

Comcast

Jan 2025 - Present

Software Developer

Cincinnati, OH

- Designed and deployed scalable microservices using Spring Boot and TensorFlow, serving 100K+ API requests daily with 99.9% uptime.
- Integrated NLP pipelines with FastAPI and Hugging Face Transformers, boosting classification accuracy by 33%.
- Optimized end-to-end model lifecycle with MLflow and Apache Airflow, reducing manual effort by 60%.
- Deployed AI APIs using Docker on AWS Lambda and SageMaker, achieving sub-300ms response latency while ensuring robust cloud infrastructure.
- Built real-time dashboards with Streamlit to visualize predictions, confidence scores, and system alerts, supporting intuitive data presentation.
- Applied transfer learning to fine-tune pretrained CNNs, improving image classification performance on limited datasets by 20%.
- Established CI/CD pipelines via GitHub Actions, streamlining deployment with a 40% reduction in release time.
- Implemented real-time monitoring with Prometheus and Grafana to track performance, latency, and model drift.
- Collaborated with frontend developers using React.js and REST APIs to deliver dynamic user interfaces and seamless data integration.
- Conducted A/B testing of AI versions, improving recommendation CTR by 11%.
- Mentored junior engineers on MLOps and deployment best practices within Agile teams, reinforcing disciplined software engineering.

Dentsu

Jan 2022 - Jul 2023

Full Stack Developer - AI Integration

Hyderabad, India

- Developed AI-integrated web applications with a Spring Boot backend and React.js frontend, enhancing fraud detection workflows.
- Designed and deployed machine learning models with Flask REST APIs and Docker, bridging full-stack development and container orchestration.
- Scheduled and monitored batch jobs using Apache Airflow, reducing manual scoring efforts by 70%.
- Utilized MongoDB and PostgreSQL to manage logs, user behaviors, and model outputs efficiently.
- Implemented secure authentication and role-based access using JWT and Spring Security, upholding information security best practices.
- Migrated legacy monolithic backend services to a microservices architecture, boosting system performance and maintainability.
- Engineered real-time dashboards with React.js and REST APIs to monitor risk scores and anomaly patterns.
- Containerized services and deployed to Kubernetes clusters, ensuring resilient cloud infrastructure.
- Authored robust test suites using JUnit and automated API testing with Postman, elevating test coverage by 45%.
- Collaborated with the DevOps team to streamline CI/CD workflows using GitHub Actions, improving deployment speed and reliability.
- Participated in Agile ceremonies and sprint planning, reinforcing disciplined software engineering practices.

Projects

Malicious URL Detection using Deep Learning

- Developed hybrid CNN+LSTM model with TF-IDF and Word2Vec features for URL phishing classification (97% accuracy).
- Reduced false positives by 18% using unsupervised clustering (KMeans) and anomaly scoring.
- Exposed models via FastAPI and deployed to Dockerized microservices with MongoDB backend for signature lookup.

Lung Cancer Detection with Bayesian Networks

- Built a diagnostic CNN classifier on X-rays (10K+ images) with 93% precision.
- Designed Bayesian inference layer to support medical triage decision-making.
- Delivered real-time results with Streamlit UI integrated into a HIPAA-compliant backend.

Smart Road Reflector with IoT + ML

- Developed IoT-based adaptive streetlight system that reduced energy use by 35%.
- Created Flask-powered API to control lighting based on traffic and ambient data.
- Proposed a city-wide rollout with evidence-based safety improvements (2x visibility).

Skills

- **Programming Languages:** Java, Python, JavaScript, C++, SQL, C#
- **Full-Stack & Web Development:** Spring Boot, FastAPI, Flask, Node.js, Express.js, React.js, HTML, CSS, Bootstrap, Full-stack Development, REST API Development, Mobile App Development, TailwindCSS, Wasm
- **AI & ML:** TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers, XGBoost, CNN, LSTM, Transfer Learning, NLP, Computer Vision
- **MLOps & Tools:** Docker, MLflow, Apache Airflow, Kubernetes, GitHub Actions, DVC, Streamlit, Gradio
- **Cloud Platforms:** AWS (EC2, S3, Lambda, SageMaker), GCP (Vertex AI, BigQuery), Azure ML
- **Databases:** MongoDB, PostgreSQL, MySQL, Microsoft SQL Server, NoSQL
- **Other Tools:** Git, GitHub, Jenkins, Prometheus, Grafana, Postman, JUnit, Agile/Scrum, SVN, Information Security

Certifications

- **Full Stack Web Development Bootcamp:**Udemy (62 hours)
Covered MERN Stack, Git, APIs, Auth, Deployment, Testing
- **MLOps Specialization:**Issued by: DeepLearning.ai (Coursera) | Year: 2024